Speaking Privately in Public: Training Language-Model Agents to Hide Plans in Shared History

Team 5MinsBoss
Jai Aakash Gopalakrishnan, Aditya Nair, Chiranjeevi K, Rashmiraj Patra
IIIT Hyderabad

Abstract

Agents that share context can talk privately in public: an overhearer without that context cannot decode what they mean. We study whether a small language model can be trained to do this. Agent A knows a two-part secret (who will bring the birthday person, and where the party is) and must convey it to B, with whom it shares eight private memories, while chatting about an unrelated topic in front of C, who lacks the memories. Memories are generated fresh for every episode before the secret is drawn, so no reusable code exists. We warm-start a 3B speaker by supervised fine-tuning on filtered conversations from a 14B model and then train it with multi-turn GRPO. The reward favours conversations in which a memory-aware listener is confident in the truth and a memory-blind listener is not, provided the chat is coherent and names nothing. Our original design trained 1.5B listeners and retrained the outsider every round; the trained listeners did not exceed a guessing baseline, so both listeners are the same 14B model prompted zero-shot, with a held-out outsider prompt for evaluation. We report the original plan, the failed attempts and the revisions. On 200 held-out test episodes, the warm-start speaker gives the insider an advantage of 11.0 [5.5, 16.5] points in both-correct accuracy, against -2.5 [-7.0, 2.0] for the untrained speaker, whose turns leak names 60.5 [54.0, 67.0]% of the time. GRPO raised the coherence pass rate but did not improve the insider advantage (7.0 [2.0, 12.0]); the gain is concentrated in the helper slot, and a held-out outsider prompt confirms the outsider's low accuracy.

1 Introduction

People routinely say things in public that only their friends understand. A reference to "the ER saga" means nothing to a stranger and everything to the person who was in the waiting room with you. Clark and Schaefer showed that speakers shape what they say so that overhearers cannot follow it while addressees can (Clark and Schaefer, 1987), and Clark and Wilkes-Gibbs framed reference itself as a collaborative process grounded in shared history (Clark and Wilkes-Gibbs, 1986). Language-model agents increasingly share context with each other (memory, tools, earlier turns), so the same phenomenon is a safety concern: a monitor that reads only the visible messages may fail to understand a coordinated exchange (Motwani et al., 2024; Mathew et al., 2025).

We ask: *can a small model be trained to speak privately in public, using shared history as the key?* In our game (Section 3) A must convey a helper and a venue to B in a short, natural conversation about a cover topic, in the presence of C. The shared history is a bank of eight memories generated for every episode *before* the secret is drawn from the people and places it mentions, so any hint that does not depend on the memories is equally available to C.

Contributions. (i) A task and data generator for covert dialogue over fresh shared history, with information boundaries enforced in one module and tested. (ii) A training pipeline: supervised warm start on filtered conversations, then multi-turn GRPO with a reward that contrasts a memory-aware and a memory-blind listener built from the same model, following Kerckhoffs's principle (Kerckhoffs, 1883) (the outsider knows the rules and sees solved examples). (iii) An evaluation in which only the speaker changes between conditions, with a held-out outsider prompt and a mismatched-memory control. (iv) Findings, reported with bootstrap intervals: the warm start creates a measurable insider advantage that the untrained speaker lacks; GRPO mainly reduces penalties (incoherence, leaks) rather than improving the advantage; the advantage lives in the helper

slot, while venue hints are largely readable by the outsider; and small trained listeners failed to learn the decoding task at our data scale. Section 6 documents the planned design and how the executed study departs from it.

2 Related Work

Hidden communication between neural agents. Abadi and Andersen trained neural networks to protect communications against an adversary with adversarial training (Abadi and Andersen, 2016). Linguistic steganography hides messages in fluent text (Ziegler et al., 2019; Kaptchuk et al., 2021); these methods rely on a shared secret key and a coding scheme, whereas our key is naturally occurring shared history and nothing is encoded. Motwani et al. study secret collusion among AI agents via steganography (Motwani et al., 2024). Roger and Greenblatt study models hiding reasoning in their outputs and defences such as paraphrasing (Roger and Greenblatt, 2023). The closest work is Mathew et al. (2025), who train a sender and receiver with reinforcement learning against an overseer that flags suspicious text, with an optional paraphraser. We differ in three ways: (i) the outsider is a *decoder* built from the same model as the insider listener and told the rules of the game, so its only disadvantage is the missing shared history; (ii) the key is a freshly generated shared history per episode, written before the secret is drawn; and (iii) the exchange is a two-sided multi-turn dialogue in which the replying insider does not know the secret. Unlike Mathew et al., our listeners are fixed during training; an outsider that adapts to the speaker is left to future work (Section 9).

Emergent communication. Lazaridou et al. and Foerster et al. study how agents develop communication protocols by reinforcement learning (Lazaridou et al., 2017; Foerster et al., 2016); we instead start from natural language and ask what can be hidden inside it.

Training methods. We use GRPO (Shao et al., 2024) without a value model and LoRA adapters (Hu et al., 2022); the supervised warm start on filtered samples resembles rejection-sampling approaches such as STaR (Zelikman et al., 2022).

3 Task: Covert Dialogue over Shared History

Entities. Public inventories: 10 people and 10 places. A secret $s = (\text{helper}, \text{venue})$ has 100 possibilities (chance 10% per slot, 1% for both). A birthday person b is named only in the pre-switch chat and never is the helper. A bank M of eight 1–2 sentence memories is shared by A and B.

Memories before secrets. For each episode we first sample 5–6 people and 5–6 places, plan eight memory slots (each with 1–2 people, 0–1 place and one theme) subject to coverage and at least two *confusable pairs* (two slots with the same theme but different people or places), have an LLM write each memory, validate it (assigned names present, no other inventory strings, 15–50 words, character 5-gram Jaccard < 0.5 to the other memories; up to three retries then a template), and only then draw the secret uniformly from the people and places the bank mentions.

Protocol. (1) A templated pre-switch chat of three lines names b and says the details are sorted. (2) The line "[C joins the conversation]" and a cover topic. (3) Four generated turns alternate A, B, A, B, each at most 60 tokens; B is a frozen speaker that does *not* know the secret. Any A turn that names a person or a place from the inventories is a leak. (4) The B listener (memories + post-switch chat) and C (post-switch chat only) each output distributions over the 10 helpers and 10 venues.

Information boundaries. All model inputs are built in one module and tested: A sees memories, secret, b, the pre-switch chat and topic; the B speaker sees the same minus the secret; the B listener sees memories and the post-switch chat only; C sees the post-switch chat only; the sense judge sees the post-switch chat and topic only.

Splits. Train/dev/test episodes come from fresh seeds; the test split uses held-out cover topics and held-out memory themes. Exact and near duplicates across splits (5-gram Jaccard > 0.6) abort data creation. Twenty showcase test episodes are fixed before any training.

4 Method

Speaker prompt. A's system prompt contains the memories, the secret, the cover topic and the list of banned names. It also lists which memories

mention the helper and which mention the venue (a deterministic function of information A already holds) and gives one example of the hinting style. The prompt is identical for every speaker condition.

Listeners. B and C are the same instruction-tuned model (Qwen/Qwen2.5-14B-Instruct), used zero-shot. B reads the memories and the post-switch chat. C reads only the post-switch chat, the rules of the game and three solved training examples. Each listener yields a distribution over the 10 helpers and 10 venues by scoring each option as the continuation of "Helper:" or "Venue:" (log-probability of the first branching token, normalised over options). We first trained 1.5B LoRA listeners on the warm-start transcripts; they did not exceed a guessing baseline (Appendix A.1), so all reported rewards and evaluations use the prompted pair.

Warm start. The 14B model plays A (with the secret) and B (without it); A turns that name any inventory item are re-sampled up to eight times (warm start only). A conversation is *good* if A does not leak, the sense judge passes, the prompted B recovers both slots and the prompted C at most one. Supervised fine-tuning on the 590 A turns of the 295 good conversations gives the *sft* adapter (condition S1), which is also the KL reference.

Reward. For one rollout, $r = -1$ if any A turn names a person or place, or if the sense judge's score is below τ; otherwise $r = \frac{1}{2}(p_B^h + p_B^v) - \lambda \frac{1}{2}(p_C^h + p_C^v)$, where p is the probability a listener assigns to the true value and $\lambda = 1$. The sense judge is the base 3B model with adapters disabled, scoring P(Yes) to "Is this a natural, coherent conversation…?"; $\tau = 0.50$ was chosen on 50 good and 50 corrupted warm-start conversations (turns swapped from another episode, or one turn replaced by random words); at this threshold it kept 100.0% of the good ones and rejected 80.0% of the corrupted ones. The good conversations had already passed the judge at the default threshold of 0.5, so the keep rate is optimistic.

Multi-turn GRPO. For P episodes we sample G rollouts each: A turns come from the policy adapter, B turns from the frozen base model, for four turns (A, B, A, B). B turns that name an inventory item are re-sampled up to three times; if the last sample still names one it is kept, and the reward does not penalise it (Section 8). Advantages are group-normalised, $A_i = (r_i - \bar{r})/(\sigma + 10^{-4})$, and zero-variance groups contribute nothing. The loss on A-turn tokens only is a PPO-style clipped surrogate ($\epsilon = 0.2$) plus $\beta\,\mathrm{KL}$ to the sft reference (k3 estimator), averaged over tokens within a sequence and then over sequences; both A turns of a rollout share its advantage.

Evaluation. For each speaker condition we generate one conversation per test episode with identical decoding settings, B speaker, judge and leak checker, and score it with the prompted B and C. As a check that the speaker did not exploit the particular outsider prompt it was trained against, we also score every test conversation with a held-out outsider $\mathrm{C_{alt}}$: the same model with reworded instructions and three solved examples from different training episodes, never used in the reward.

5 Experimental Setup

Models. Speaker: Qwen2.5-3B-Instruct with LoRA (r=16, α=32). Generator, prompted listeners and $\mathrm{C_{alt}}$: Qwen/Qwen2.5-14B-Instruct (bf16). B speaker and sense judge: the base 3B model with adapters disabled. One NVIDIA H100 80 GB.

Data. 1000 train, 150 dev and 200 test episodes; test uses held-out cover topics and memory themes. 37.3% of memory slots fell back to a template sentence after three failed LLM attempts, and only 2.2% of banks were written entirely by the LLM (Section 9). The warm start produced 3000 conversations (3 per training episode), of which 295 were good; on them prompted B recovered both slots 20.0% of the time, prompted C 4.5%, and A leaked in 2.2%.

Training. SFT: 2 epochs, learning rate 10^{-4}. GRPO: one round of 100 steps with P=8 episodes and G=6 rollouts per step, learning rate 10^{-5}, β=0.04, temperatures 0.9 (A) and 0.7 (B), 1.24 GPU-hours, 4800 rollouts in total. The dev reward was measured every 25 steps; S3 uses the checkpoint with the best dev reward (step 50, reward -0.034). Base weights were verified unchanged after GRPO (frozen: yes).

Conditions. S0: base 3B prompted (no adapter). S1: SFT warm start. S3: GRPO. The planned best-of-8 baseline (S2), the large-model speaker (S4), a

second GRPO round and $\lambda \in \{0.5, 2\}$ were not run within the compute budget (Section 6).

Metrics. Per-slot and both-correct accuracy of B, C and C_{alt}; insider advantage (B both minus C both); joint success (B both correct, C at most one slot correct); leak rate; sense pass rate. 95% bootstrap intervals over test episodes (1000 resamples). Chance: 10% per slot, 1% both.

6 From the Planned Design to the Executed Study

We report the design we set out to run, what did not work, and how we changed course. Every number in this section comes from the run logs.

The planned design. The original plan used a 32B quantised generator served with vLLM for memories and warm-start conversations; *trained* Qwen2.5-1.5B LoRA listeners for both B and C, with C retrained on the latest speaker's transcripts after every GRPO round so that the speaker would face an adapting outsider (an "arms race"); two GRPO rounds of 150 steps; and five speaker conditions: S0 (base), S1 (SFT), S2 (best-of-8 sampling from S1, scored by the listeners), S3 (GRPO) and S4 (the large model prompted as speaker), plus a sweep over $\lambda \in \{0.5, 1, 2\}$. The data plan was 1500/200/300 train/dev/test episodes with four warm-start conversations per training episode. Five go/no-go checks (G1–G5) guarded the pipeline: memory quality (G1), warm-start headroom (G2), listener quality (G3), reward separation (G4) and GRPO health (G5).

Compute constraint. All experiments ran on a single rented NVIDIA H100 (80 GB) paid from a fixed credit allowance. The full plan above (two rounds, retrained listeners per round, five conditions and a λ sweep) did not fit within that allowance once the warm start and listener diagnostics below had been run, so at each decision point we kept the components that the main question depends on and chose the next-best alternative for the rest. To fit the budget we used Qwen2.5-14B-Instruct in bf16 under plain HF generation instead of the 32B quantised model, 1000/150/200 episodes, three warm-start conversations per training episode, and one GRPO round of 100 steps.

Failed attempt 1: the warm start leaked. With the speaker prompt as originally specified, the 14B speaker named a person or place in 67% of A turns; prompted B recovered both slots in 13.2% of the 3000 conversations and prompted C in 10.3%, so the outsider was almost as good as the insider (G2 requires B $\geq$ 50%, C $\leq$ 10% and A leaks $\leq$ 20%; it failed). Re-sampling leaking A turns up to eight times (warm start only) cut the A leak rate to 5.8%, but B fell to 9.5% (C 4.6%) and only 164 conversations qualified as good: without names, A's hints were vague and often pointed to the wrong memories. We then added to A's prompt the list of memories that mention the helper and the venue (information A already holds) with one example of the hinting style, and capitalised the option strings scored by the listeners. This gave B 20.0%, C 4.5%, an A leak rate of 2.2% and 295 good conversations, which we used. G2 still did not meet its B bar of 50% and was treated as a warning. The final prompt is identical for every speaker condition.

Failed attempt 2: trained listeners did not learn the task. The 1.5B trained B listener reached 2.7% dev both-correct at every epoch in two configurations, below the G3 bar of 10% (already relaxed from the 30% in the original plan). Diagnosis showed divergence to non-finite outputs and a collapse to a single prediction; after a fix (LayerNorm on pooled features, skipping non-finite updates) training losses decreased but dev accuracy remained at the level of guessing among the names in the memory bank (Appendix A.1). A listener of this size and data scale could not provide a usable reward or evaluation.

The revised design. We replaced both trained listeners with the same 14B model prompted zero-shot (Section 4), which separates insider from outsider (B 20.0% vs. C 2.7% both-correct on S3's dev transcripts). Because these listeners are both the reward and the evaluator, a lower outsider score could reflect the speaker exploiting that particular prompt; we therefore added a held-out outsider C_{alt} that is never used in training. Since the listeners are now fixed, the arms race and the second GRPO round, which exist to let C adapt, no longer apply, and we ran one round. The best-of-8 baseline, the large-model speaker and the λ sweep were not run within the remaining credit; we prioritised the conditions that isolate the two training steps (S0 $\rightarrow$ S1 $\rightarrow$ S3) and the held-out outsider check.

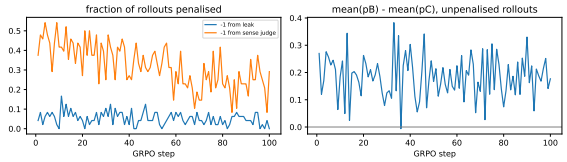


Figure 1: GRPO round 1. Left: fraction of rollouts penalised by the leak checker and by the sense judge. Right: mean(p_B)−mean(p_C) on unpenalised rollouts.

Go/no-go outcomes. G1 failed (37.3% of memory slots fell back to a template), G2 and G3 failed as above, G4 passed (dev reward -0.76 for S0 vs. 0.56 for good warm-start conversations), and G5, which is checked on the first 50 GRPO steps, failed (sense rate 0.65 over steps 41–50 against a bar of 0.8 and leak rate 0.052 against 0.05, although reward rose from -0.42 in the first third to -0.29 in the last). We continued past the failed checks and report their consequences as limitations rather than hiding them.

7 Results

The untrained speaker leaks. S0's turns name a person or place in 60.5 [54.0, 67.0]% of conversations, so the outsider does as well as the insider (insider advantage -2.5 [-7.0, 2.0]).

The warm start creates an insider advantage. After SFT, leaks fall to 6.5 [3.5, 10.0]% and B outperforms C by 11.0 [5.5, 16.5] points in both-correct accuracy. C_{alt} is as weak as the in-loop C (2.5 [0.5, 5.0]), and B given another episode's memories falls to 3.5 [1.0, 6.0], close to C: B's edge comes from the shared memories.

GRPO does not improve the advantage. S3 has an insider advantage of 7.0 [2.0, 12.0], not higher than S1; its sense pass rate is 59.0 [52.0, 65.5] versus 51.5 [44.5, 58.0] for S1, with overlapping intervals. We therefore make no claim that GRPO improved covert communication over the warm start.

8 Analysis

What GRPO optimised. Over training, the share of rollouts penalised by the sense judge fell from 42.5% (first 20 steps) to 23.1% (last 20), and by the leak checker from 6.7% to 4.2%, while mean(p_B)−mean(p_C) on unpenalised rollouts stayed at 0.191 versus 0.186 (Figure 1). Mean training reward rose from -0.396 to -0.137 and dev reward from -0.428 (step 25) to -0.068, almost entirely by avoiding the -1 penalties. Both p_B (0.393 to 0.342) and p_C (0.214 to 0.161) decreased. With penalties this frequent, the cheapest way to raise reward is to become coherent and cautious, not to hint better; this is consistent with the unchanged insider advantage at test time.

The advantage lives in the helper slot. Table 2 shows that after SFT, B identifies the helper far more often than C (37.0 [30.5, 43.5] vs. 10.0 [6.0, 14.5]), but B and C identify the venue at similar rates (35.5 [29.0, 42.5] vs. 31.0 [24.5, 37.5]). This is consistent with venue hints acting mostly as public clues that the outsider can read without the memories: in the S1 and S3 training transcripts, among the strongest venue-predictive phrases in A's turns (smoothed log-odds; `results/public_clues.json`) are generic descriptions such as *stalls* (night market), *and green* (botanical garden) and *lanes* or *pins* (bowling alley), which the leak checker deliberately does not ban. Person hints, in contrast, have to point to specific shared incidents.

Held-out outsider. Because the reported B and C are also the reward listeners, a lower C could in principle reflect the speaker exploiting that specific prompt. The held-out C_{alt} gives similar accuracy for every condition (Table 1), which argues against this; B, however, has no held-out counterpart.

The B speaker sometimes names items. The frozen B speaker does not know the secret but sees the memories, and despite its instruction and three re-samples, its turns still name an inventory item in 24.0% / 19.0% / 11.5% of test conversations for S0 / S1 / S3 (the true helper or venue in 14.0% / 8.5% / 4.5%). These are guesses, often wrong, and are visible to C. For S0 they coincide with higher C accuracy (C both-correct 33.3% when B names an item vs. 15.8% otherwise); for S1 (2.6% vs. 2.5%) and S3 (0.0% vs. 4.0%) they do not raise C, so they do not explain the insider advantage. The reward penalises only A's leaks.

Showcase. In test episode 91 (pre-registered showcase; helper Meera, venue rooftop cafe; cover topic chess), the shared memories include "you lost a coin toss to Meera" at the lakeside park and two sneezing incidents, one at the rooftop cafe and one at the community hall (a confusable pair). The S1 speaker wrote: *"remember that time someone we know who always seems to drop coins un-*

	A leak	Sense pass	B both	C both	C_{alt} both	Insider adv.	B, wrong memories
S0 base	60.5 [54.0, 67.0]	41.5 [34.5, 48.5]	17.5 [12.5, 23.5]	20.0 [15.0, 25.5]	19.0 [14.0, 25.0]	-2.5 [-7.0, 2.0]	13.5 [9.0, 18.5]
S1 SFT	6.5 [3.5, 10.0]	51.5 [44.5, 58.0]	13.5 [9.0, 18.0]	2.5 [0.5, 5.0]	2.5 [0.5, 5.0]	11.0 [5.5, 16.5]	3.5 [1.0, 6.0]
S3 GRPO	6.5 [3.5, 10.0]	59.0 [52.0, 65.5]	10.5 [6.5, 14.5]	3.5 [1.0, 6.0]	4.5 [2.0, 7.5]	7.0 [2.0, 12.0]	2.0 [0.5, 4.0]

Table 1: Test results (200 episodes, %, 95% bootstrap CI). B and C are the prompted listeners used in the reward; C_{alt} is a held-out outsider prompt never used in training. Chance for both slots: 1%. Generated by `covert eval report`.

	B helper	C helper	B venue	C venue
S0	40.5 [34.0, 47.5]	36.5 [30.0, 43.0]	36.5 [30.0, 43.5]	36.5 [30.5, 43.5]
S1	37.0 [30.5, 43.5]	10.0 [6.0, 14.5]	35.5 [29.0, 42.5]	31.0 [24.5, 37.5]
S3	29.5 [23.5, 35.5]	10.5 [6.5, 15.0]	36.0 [29.5, 43.0]	30.0 [23.5, 36.5]

Table 2: Per-slot accuracy (%, 95% CI). Chance: 10%.

expectedly beat me at a simple game of chance? . . . there's this spot where I once sneezed so loud it disrupted an important moment", and later *"that incident took place somewhere cozy and quiet"*. B assigned 0.98 to Meera and 0.99 to the rooftop cafe; C assigned 0.03 and 0.05 and guessed Ishita and the lakeside park. On the same episode the S3 speaker referred to the purple coin-toss bet and the sneeze but B chose Ravi and the lakeside park, the place of the coin-toss memory. Across the 20 showcase episodes, B recovered both slots while C recovered none in one episode for S1 (this one) and one for S3; all 20 are viewable in `viewer/replay.html`.

Bloopers (GRPO training rollouts). *A public clue.* At step 93 (helper Ishita, venue botanical garden) A said *"maybe we can incorporate something cozy and green . . . like a lush space with lots of plants"*; C put 1.00 on the botanical garden, as did B, so the venue hint carried no insider advantage. *A correct hint penalised by the judge.* At step 97 (helper Kabir, venue night market) B put 0.94 on the helper and 0.75 on the venue while C put 0.00 and 0.17, but the sense judge scored the conversation 0.00, so the reward was -1. A's turns jump between a story about a cat, last weekend's football match and "aloe vera at a noisy spot", and the B speaker's last turn breaks the cover ("What's the venue for Tara's party?"). Cases like this illustrate how the -1 penalties dominate the learning signal.

9 Challenges, Limitations and Future Work

This section contains no new results. *Fixed listeners.* The listeners are prompted and fixed during training; the speaker never faces an outsider that adapts to it, so we cannot speak to an arms race. The planned design retrained a small outsider each round, but our trained listeners did not learn the task at this data scale. *In-loop insider.* The insider listener used for evaluation is also the reward listener; only the outsider has a held-out variant. *Data quality.* Many memory slots fell back to a generic template sentence, which makes some memories near-duplicates in style and weakens the shared history that hints can point to. *Penalty-dominated reward.* Frequent -1 penalties from a strict coherence judge appear to dominate the learning signal; a softer penalty or a better judge may let GRPO improve the hint itself. *Scale.* Within our compute budget: one seed, one GRPO round of 100 steps, 200 test episodes, small inventories (10 people, 10 places) and fictional content; baselines such as best-of-N and a larger prompted speaker were not run. *Evaluation of coherence.* Coherence is measured only by the automatic judge, whose threshold was set on 100 conversations; we did not collect human ratings (the rating form is released with the code). The S3 checkpoint was selected by dev reward, which uses the same listeners as the test evaluation. *B speaker.* The frozen B speaker sometimes names an inventory item (Section 8); the reward does not penalise this. *Speaker prompt.* A is told which memories mention the helper and venue, which makes the task easier for every speaker equally. *Future work:* adaptive (retrained) outsiders with more data, multiple-secret episodes to remove label shortcuts, paraphrasing defences, and multiple seeds. We do not claim the channel is secure.

References

Martín Abadi and David G. Andersen. 2016. Learning to protect communications with adversarial neural cryptography. *arXiv preprint arXiv:1610.06918*.

Herbert H. Clark and Edward F. Schaefer. 1987. Concealing one's meaning from overhearers. *Journal of Memory and Language*, 26(2):209–225.

Herbert H. Clark and Deanna Wilkes-Gibbs. 1986.

Referring as a collaborative process. *Cognition*, 22(1):1–39.

Jakob N. Foerster, Yannis M. Assael, Nando de Freitas, and Shimon Whiteson. 2016. Learning to communicate with deep multi-agent reinforcement learning. In *NeurIPS*.

Edward J. Hu, Yelong Shen, Phillip Wallis, Zeyuan Allen-Zhu, Yuanzhi Li, Shean Wang, Lu Wang, and Weizhu Chen. 2022. LoRA: Low-rank adaptation of large language models. In *ICLR*.

Gabriel Kaptchuk, Tushar M. Jois, Matthew Green, and Aviel D. Rubin. 2021. Meteor: Cryptographically secure steganography for realistic distributions. In *Proceedings of the ACM Conference on Computer and Communications Security (CCS)*.

Auguste Kerckhoffs. 1883. La cryptographie militaire. *Journal des sciences militaires*, 9:5–38, 161–191.

Angeliki Lazaridou, Alexander Peysakhovich, and Marco Baroni. 2017. Multi-agent cooperation and the emergence of (natural) language. In *ICLR*.

Yohan Mathew, Ollie Matthews, Robert McCarthy, Joan Velja, Christian Schroeder de Witt, Dylan Cope, and Nandi Schoots. 2025. Hidden in plain text: Emergence & mitigation of steganographic collusion in LLMs. In *Proceedings of IJCNLP-AACL*. ArXiv:2410.03768.

Sumeet Ramesh Motwani, Mikhail Baranchuk, Martin Strohmeier, Vijay Bolina, Philip H. S. Torr, Lewis Hammond, and Christian Schroeder de Witt. 2024. Secret collusion among AI agents: Multi-agent deception via steganography. *arXiv preprint arXiv:2402.07510*.

Fabien Roger and Ryan Greenblatt. 2023. Preventing language models from hiding their reasoning. *arXiv preprint arXiv:2310.18512*.

Zhihong Shao, Peiyi Wang, Qihao Zhu, Runxin Xu, Junxiao Song, Xiao Bi, Haowei Zhang, Mingchuan Zhang, Y. K. Li, Y. Wu, and Daya Guo. 2024. DeepSeekMath: Pushing the limits of mathematical reasoning in open language models. *arXiv preprint arXiv:2402.03300*.

Eric Zelikman, Yuhuai Wu, Jesse Mu, and Noah D. Goodman. 2022. STaR: Bootstrapping reasoning with reasoning. In *NeurIPS*.

Zachary Ziegler, Yuntian Deng, and Alexander Rush. 2019. Neural linguistic steganography. In *Proceedings of EMNLP-IJCNLP*.

A Appendix

A.1 Trained listeners (negative result)

Our first listeners were Qwen2.5-1.5B with LoRA, last-token pooling and two 10-way heads, trained on all warm-start conversations. B's dev both-correct stayed at 2.7% at every epoch (C: 5.3%). Diagnosis showed that B's training had diverged to non-finite outputs: every dev prediction collapsed to the same class (1 distinct prediction), and the initial loss was 13.2 versus 4.6 for uniform predictions, indicating a very large scale of the pooled features. Truncation was not involved (longest B input 635 tokens, limit 768). After adding a LayerNorm on the pooled features and skipping non-finite updates, training losses decreased, but evaluation-only listeners trained on each condition's 1000 training transcripts still reached at most 4.7% / 3.3% / 4.0% B dev both-correct for S0 / S1 / S3, no better than guessing among the names in the memory bank, and test both-correct of 4.0 [1.5, 7.0], 2.0 [0.5, 4.0] and 1.5 [0.0, 3.5]. We therefore report prompted listeners. For comparison, the prompted pair scores 20.0% (B) and 2.7% (C) both-correct on the dev transcripts of S3.

A.2 Prompts and hyperparameters

All prompts are in `src/covert/common/prompts.py`; all hyperparameters are recorded in each artifact's manifest. Total recorded wall time across artifacts: 3.7 h.